

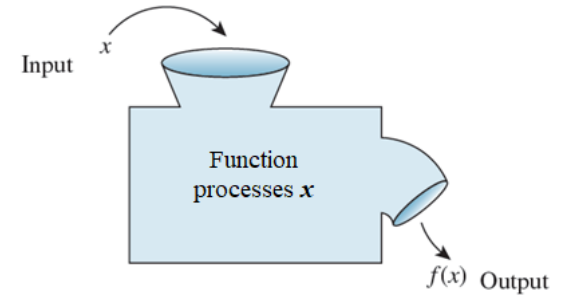
Understanding Functions

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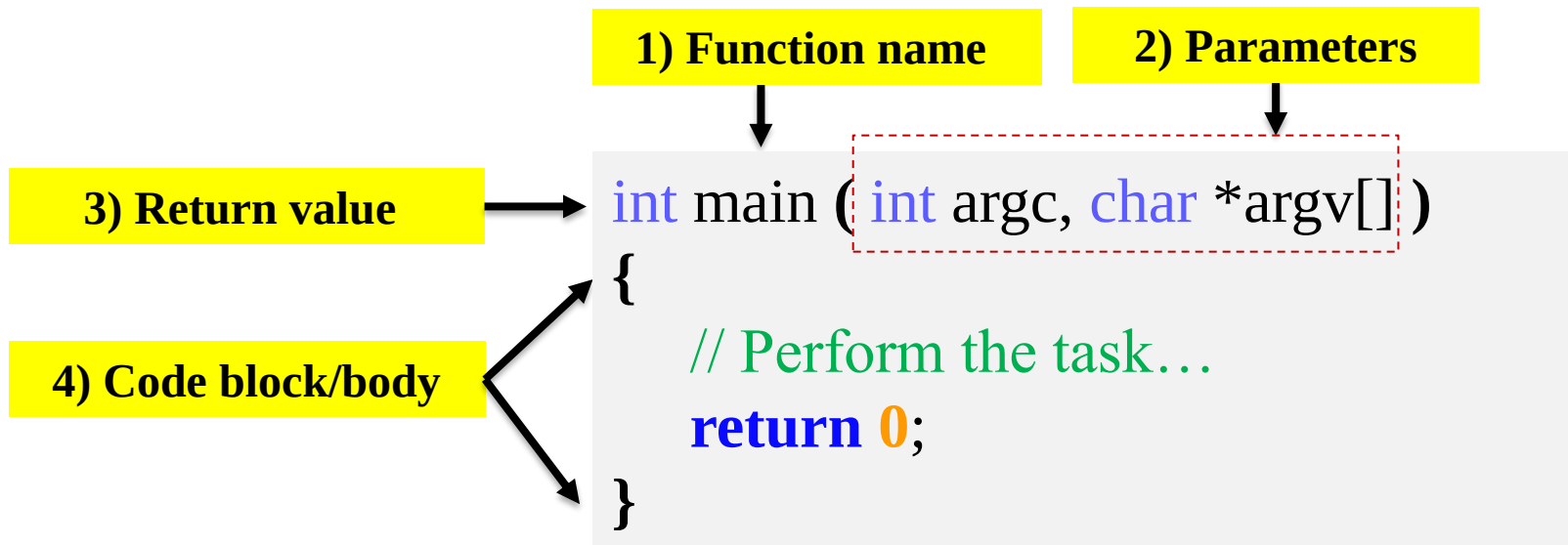
4/01/2022

What is a function?

👉 A function is a block of code that performs a specific task.



👉 A typical C++ function declaration:



An example of a C++ program

```
5 // Function declaration
6 void swap(int &x, int &y);
7 void print(int x, int y);
8
9 // The main function can call other functions in the program.
10 int main() {
11     // Local variables declaration
12     int n1 = 10;
13     int n2 = 20;
14
15     // Call function print to display numbers
16     print(n1, n2);
17
18     // Call function swap to swap two numbers
19     swap(n1, n2);
20
21     // Display numbers after swap
22     cout << n1 << ", " << n2 << endl;
23
24     return 0;
25 }
```

Diagram illustrating function calls from the main function:

- A red dashed arrow points from the call `print(n1, n2);` (line 16) to the definition of the `print` function (lines 5-7).
- A red dashed arrow points from the call `swap(n1, n2);` (line 19) to the definition of the `swap` function (lines 6-7).

Function definitions shown in callouts:

print function:

```
// Output display: x,y
void print(int x, int y) {
    cout << x << ", " << y << endl;
}
```

swap function:

```
// Swap the content of two numbers
// at the address of x and address y
void swap(int &x, int &y) {
    int z = x;
    x = y;
    y = z;
}
```

Run a C++ program

```
5 // Function declaration
6 void swap(int &x, int &y);
7 void print(int x, int y);
8
9 // The main function can call other functions in the program.
10 int main() {
11     // Local variables declaration
12     int n1 = 10;
13     int n2 = 20;
14
15     // Call function print to display numbers
16     print(n1, n2);
17
18     // Call function swap to swap two numbers
19     swap(n1, n2);
20
21     // Display numbers after swap
22     cout << n1 << ", " << n2 << endl;
23
24     return 0;
25 }
```

Compile

Executable
program

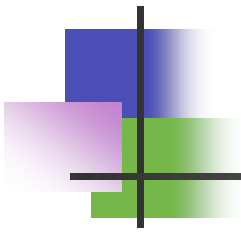
Command Prompt

```
C:\Users\jpham>demo.exe
10,20
20,10
```

Main Memory

...	...
	...0xFFFFFFFF
int y	...0x00000040
int x	...0x00000034
void print (int x, int y)	...0x00000040
void swap (int &x, int &y)	...0x00000028
...	...
cout << n1 << ", " << n2 << endl	...0x00000020
Call swap(&n1, &n2)	...0x00000016
Call print(10, 20)	...0x00000012
int n2 = 20	... 0x00000008
int n1 = 10	... 0x00000004
Call int main()	0x00000001
	0x00000000

A
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Summary

- To define a function, the function's return type, name, and has zero or more parameters.
- A function is a block of code that can be called by other statements.
- To call a function, the name of the function, and the parameters.
- The parameters can be passed by value or by reference.
- The `main()` function can call other functions in the program.

Questions

